

Mathematics for Machine Learning:

Homework 10

1. Calculate the integral.

a) $\iint_D x \sin(x+y) dx dy, D = [0, \pi] \times \left[0, \frac{\pi}{2}\right],$

b) $\iint_D xy^2 dx dy, D = \{(x, y) : 0 \leq x \leq 1, x^3 \leq y \leq x^2\},$

c) $\iint_D \frac{x^2}{y^2+1} dx dy$, where D is the domain bounded by the curves

$$y = x, y = 0, xy = 1, x = 2.$$

2. Let X be a uniformly distributed random variable on the interval $(0, 1)$. Find the correlation of the following random variables

a) X and X^2 ,

b) X and X^3 .

3. Let $X_n, n \in \mathbb{N}$ be a sequence of random variables such that their distribution function has the following forms

$$F_{X_n}(\omega) = \begin{cases} 1 - \left(1 - \frac{1}{n}\right)^{nx} & , \text{ if } x > 0 \\ 0, & \text{ if } x \leq 0. \end{cases}$$

Show that X_n converges in distribution to the exponential random variable with $\lambda = 1$.

4. Let $\lambda > 0$ and $X_n, n \in \mathbb{N}$ be a sequence of random variables such that

$$X_n \sim \mathbf{Binomial}\left(n, \frac{\lambda}{n}\right)$$

for $n > \lambda$. Show that X_n converges in distribution to **Poisson**(λ).

5. Let $X_n, n \in \mathbb{N}$ be a sequence of random variables such that

$$X_n(\omega) = \begin{cases} n^2, & \text{with probability } \frac{1}{n}, \\ 0, & \text{with probability } 1 - \frac{1}{n}. \end{cases}$$

Find the limit of X_n in means of the probability convergence.

6. Let X_n be a uniform random variable on the interval $\left(0, \frac{1}{n}\right)$, for all $n \in \mathbb{N}$.

Find the limit of X_n in means of the r -th mean, where $r \geq 1$.

7. Let joint PDF of random variables X and Y is

$$f(x, y) = \frac{a}{1 + x^2 + y^2 + x^2 y^2}.$$

- (a) Calculate the coefficient a .
- (b) Calculate joint CDF of random variables X and Y .
- (c) Calculate marginal CDFs of random variables X and Y .
- (d) Are X and Y independent?

8. Let joint PDF of random variables X and Y is

$$f(x, y) = \begin{cases} a \left(x^2 + \frac{xy}{2} \right), & \text{if } (x, y) \in [0, 1] \times [0, 2] \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Calculate the coefficient a .
- (b) Calculate $\mathbb{P}(X > Y)$.
- (c) Calculate $\mathbb{P}(X > 0.5 | Y < 0.5)$.